

Python Capstone Project 2

Smart Cricket Analytics System

Business Scenario

A cricket league wants insights about:

```
Players
Teams
Runs
Strike Rate
Boundaries
Match Performance
Team Statistics
```

The analytics team receives player statistics in CSV format and needs reports.

Input File

players.csv

```
player_id,player_name,team,matches,runs,fours,sixes
101,Virat Kohli,RCB,14,741,62,38
102,Rohit Sharma,MI,14,560,48,25
103,Shubman Gill,GT,14,620,55,28
104,KL Rahul,LSG,14,590,50,22
105,Ruturaj Gaikwad,CSK,14,610,58,24
106,Hardik Pandya,MI,14,430,30,20
107,MS Dhoni,CSK,14,280,18,16
108,Faf du Plessis,RCB,14,670,60,32
109,Sanju Samson,RR,14,550,45,25
110,Yashasvi Jaiswal,RR,14,625,57,30
```

```
111,Rinku Singh,KKR,14,480,35,29
112,Andre Russell,KKR,14,450,28,35
```

Part 1 – File Handling

Task 1

Read players.csv.

Task 2

Display all records.

Task 3

Count total players.

Part 2 – Player Analytics

Task 4

Find highest run scorer.

Task 5

Find lowest run scorer.

Task 6

Calculate average runs.

Task 7

Display players scoring more than 600 runs.

Task 8

Display players scoring less than 500 runs.

Part 3 – Team Analytics

Task 9

Count players by team.

Task 10

Calculate total runs by team.

Task 11

Find team with highest runs.

Task 12

Find team with lowest runs.

Part 4 – Boundary Analysis

Task 13

Find player with most fours.

Task 14

Find player with most sixes.

Task 15

Calculate total fours hit in tournament.

Task 16

Calculate total sixes hit in tournament.

Part 5 – Lists, Sets and Dictionaries

Task 17

Store all player names in a list.

Sort alphabetically.

Task 18

Store all teams in a set.

Display unique teams.

Task 19

Create dictionary:

```
{  
    team : total_runs  
}
```

Task 20

Create dictionary:

```
{  
    player_name : runs  
}
```

Part 6 – Functions

Create functions:

Task 21

```
find_top_scorer()
```

Task 22

```
calculate_average_runs()
```

Task 23

```
find_best_team()
```

Task 24

```
find_total_boundaries()
```

Part 7 – Exception Handling

Task 25

Handle missing CSV file.

Task 26

Handle invalid run values.

Task 27

Handle invalid match counts.

Part 8 – NumPy

Create NumPy array of runs.

Task 28

Find:

```
Total Runs  
Average Runs  
Maximum Runs  
Minimum Runs  
Standard Deviation  
Median
```

Part 9 – Pandas

Task 29

Read CSV using Pandas.

Task 30

Display top 5 run scorers.

Task 31

Display players sorted by runs descending.

Task 32

Group by team and calculate total runs.

Task 33

Group by team and calculate average runs.

Task 34

Display players with runs > 600.

Task 35

Find top team.

Report Generation

Generate:

```
cricket_report.txt
```

Containing:

```
Total Players
Total Runs
Average Runs
Highest Scorer
Lowest Scorer
```

```
Team Wise Runs
```

Top 5 Players

Most Fours

Most Sixes

Bonus Tasks

Task 36

Generate:

```
top_players.csv
```

Players with runs above 600.

Task 37

Generate:

```
team_summary.csv
```

Containing:

```
Team
```

```
Total Runs
```

```
Average Runs
```

```
Player Count
```

Task 38

Create menu-driven application.

```
1. Player Analysis
```

```
2. Team Analysis
```

```
3. Boundary Analysis
```

4. Export Reports

5. Exit
